Content-Aware Cross-Platform Engagement Forecasting and Real-Time LLM Recommendations for Social Media Creators

V. Mahamadh Basha, L. Rama Kasturi, K. Mythili, T. Reddaiah, and [Project Guide]
Department of Computer Science and Engineering, K.S.R.M. College of Engineering, Andhra Pradesh, India
Email: [CONFIRM e-mail addresses]

***Abstract*—Creators who publish the same idea as an Instagram post, a YouTube Short and a longer video must read separate analytics dashboards that report numbers but say little about the content behind them. This paper describes a practical decision-support system that tries to connect the content of a post, the platform it appeared on, the creator's own history and the audience response. Posts and engagement snapshots are collected from YouTube and Instagram through their supported APIs and stored in a common record that keeps platform-specific fields. Text descriptors, basic visual descriptors and, when enabled, CLIP embeddings and Whisper transcripts describe the content. Engagement forecasting is a supporting module. An event-driven workflow reacts to new uploads, under-performance and comment surges, retrieves similar earlier posts by the same creator, and asks a large language model for suggestions that are checked for grounding, with a rule-based fallback. We report the architecture, which components are optional or incomplete, the stored records of one inspected deployment, and an evaluation protocol. The current evidence comes from a single YouTube account, so no cross-platform, forecasting or recommendation performance is claimed. [REQUIRED: measured MAE, Spearman correlation and recommendation rubric scores]**

***Index Terms*—cross-platform social media analysis, content-aware computing, multimodal content understanding, engagement forecasting, large language models, retrieval-augmented generation, event-driven analytics**

I. Introduction

A creator may post one idea in several forms: an Instagram Reel, a YouTube Short, a long YouTube video, or an image with a caption. The formats, available metrics and audience behaviour differ by platform, so the creator reads several dashboards and decides by hand what to change. These dashboards report views, likes and watch time. They do not say which characteristics of the content, such as the caption, the opening of a video or the call to action, may have contributed.

Problem. The question studied here is how content from different platforms can be represented in a common form, combined with platform-specific and creator-specific context, and then used to give recommendations that point to evidence.

Gap. Published work that we examined predicts popularity or engagement for a platform or dataset [1]–[3], studies transfer between domains [4], or retrieves video segments to answer queries [5], [6]. Within this set we did not find a workflow that links a common cross-platform post representation, content descriptors, a creator's own history, event triggers and retrieval-supported suggestions. The statement covers only the papers examined and is not a claim that no such system exists.

Objectives. (1) Collect posts and engagement snapshots from YouTube and Instagram into one schema. (2) Extract content descriptors beyond engagement numbers. (3) Estimate performance relative to the creator's own history. (4) Detect new-upload, under-performance and comment-surge events. (5) Generate suggestions from retrieved similar posts and check that they are grounded. (6) Present results in one dashboard.

Contributions. We claim only the following, and only as implemented features, not as measured improvements:

- A common post record that standardises structure but keeps platform-specific fields and metric meanings (Section III-C).
- Content-aware descriptors with explicit handling of missing modalities and optional deep encoders (Section III-D).
- An event-driven workflow that retrieves similar historical posts of the same creator and applies grounding checks and a fallback before showing advice (Sections III-G to III-I).
- An honest account of what is implemented, optional, incomplete and still unevaluated (Sections IV–VI).

Forecasting supports this workflow and is not the main contribution.

II. Related Work

A. Analytics and engagement measurement

Watch-based measures are less biased than raw view counts [7]. For short videos, the engagement continuation rate (ECR) and the normalised average watch percentage were proposed with the SnapUGC dataset, and quality scores from older video-quality datasets were found to correlate weakly with engagement [1]. A later challenge compared many submissions on ECR [8]. These works define what to predict; they are not creator-facing tools.

B. Cross-platform and cross-domain analysis

Roy et al. transfer knowledge from social streams to improve video popularity prediction, using Twitter and YouTube data [4]. Riis et al. study one platform, Instagram, with a ranking formulation and an ablation of feature sets [3]. We read both at abstract level only. Neither, in the form we examined, keeps platform-specific metric meanings in a shared record, and cross-platform content representation remains thinly covered in the set we examined.

C. Multimodal content understanding

CLIP [9] and Whisper [10] are the pretrained encoders behind the optional visual and speech modules here. Text-conditioned attention has been used to fuse video and text for quality assessment [11], and fine-tuned multimodal language models led the engagement challenge [12], [13]. Sun et al. found that audio input helped [13]. These models are large; our default pipeline uses lighter descriptors, so the comparison is one of design, not of accuracy.

D. Popularity and engagement prediction

SMTPD gives 30-day popularity sequences for 282K multilingual YouTube posts and shows that early popularity dominates later accuracy [2]. Gradient-boosted trees such as LightGBM are strong tabular baselines [14]. Reported scores are benchmark results on their own data and cannot be compared with ours.

E. Recommendation and retrieval-augmented generation

RAG conditions a generator on retrieved text [15], usually with dense retrieval [16]. VRAG retrieves, re-ranks and filters video segments before question answering [5]; CadenceRAG decomposes long queries and aligns sub-queries in time [6]; ModaRoute lets an LLM choose which modality index to search [17]; MissRAG retrieves a substitute for a missing modality [18]. All retrieve from content corpora, not from a creator's history to support advice. MissRAG is relevant because YouTube and Instagram expose different fields.

F. Event-driven decision support

We have not yet added verified peer-reviewed sources for event-driven analytics. *[CONFIRM: add 2 verified papers on event detection or stream processing]*

G. Comparison and research gap

Table I compares the works. None of them, as examined, combines (a) a record that keeps platform-specific fields, (b) content descriptors, (c) the creator's own history, (d) event triggers and (e) retrieval-supported suggestions with a grounding check. This combination is the focus of the present work.

III. Proposed Methodology

A. Overall architecture

Fig. 1 shows the workflow. A creator connects accounts; connectors collect posts and engagement snapshots; records are normalised; content features are extracted; history and a supporting forecast give performance context; events trigger retrieval and an LLM-assisted recommendation; grounding checks decide whether the advice or a fallback is stored and shown on the dashboard.

B. Cross-platform data collection

The YouTube connector uses the YouTube Data API v3 for video and channel information, with scheduled synchronisation and, where supported, push notifications. The Instagram connector uses the Instagram API with Instagram Login for media and account information, with webhooks where configured. Available fields depend on permissions, account type and API access, so not every metric is available for every account. Whether both connectors work end to end must be verified before claiming live operation on both platforms.

C. Unified post representation

Each post becomes one record (Table II). The schema standardises structure only. Views, impressions, reach, watch time and interactions can mean different things on each platform, so engagement is stored as time-stamped snapshots together with the original platform metadata, and the system does not assume that two raw values from different platforms are equal in meaning.

D. Content feature extraction

Fig. 2 shows the content path. *Text* descriptors include caption length, hashtag count, emoji use, question wording, presence of a call to action and posting day and time, together with a text embedding; the default configuration uses hash-based embeddings. *Visual* information has two levels: basic descriptors such as brightness and colourfulness, which need no deep model, and optional CLIP embeddings [9]. *Speech* is optional: Whisper [10] produces a transcript, which is then analysed as text. A post without a modality, for example an image post without speech, is treated as having no such information and not as a failure. The logical representation is therefore text features plus available visual features plus available speech features plus platform context.

E. Cross-platform content awareness

The system keeps three kinds of information apart: shared content information (topic, caption, visual and speech information, call to action), platform-specific information (format, available metrics, publishing fields) and creator-specific information (earlier posts, history, publishing habits). This allows two posts on different platforms that carry the same idea to be compared on content, while their audience responses stay in platform terms. This is an architectural capability. A real cross-platform experiment needs enough data from both platforms and a comparison protocol (Section V).

TABLE I
COMPARISON OF EXAMINED WORKS (AS DESCRIBED IN THE CITED PAPERS)

Ref.	Problem	Method	Data	Limitation for this work
[4]	Sudden popularity bursts of video	Transfer learning from social streams to the video domain	Tweets and YouTube videos	Topic-level signal; no creator-specific advice
[3]	Instagram popularity	Transferred visual semantics, ranking, ablation	Instagram posts	One platform; population model
[1], [8]	Engagement of new short videos	Content-only multimodal features, cross-modal attention	SnapUGC (Snapchat)	No creator history, explanation or action
[2]	30-day popularity sequence	ResNet-101, multilingual BERT, LSTM	SMTPD, YouTube	Needs day-one popularity; no advice
[12], [13]	Engagement continuation rate	Fine-tuned multimodal language models, ensembles	SnapUGC challenge	Heavy compute; score only
[11]	Quality of AI-generated video	Text-conditioned cross-modal attention	Challenge videos	Different task
[15], [16]	Knowledge-intensive text tasks	Dense retrieval plus generator	Text corpora	Not social media or creator data
[5], [6]	Long-video question answering and search	Retrieve, re-rank, filter, multimodal LLM	Vimeo video collections	Retrieves video, not creator history
[17]	Cost of multimodal retrieval	LLM picks which modality index to search	Large clip collection	Not an engagement task
[18]	Missing modalities in language models	Retrieve substitute modality, prompt change	Five datasets	Zero-shot; no engagement task

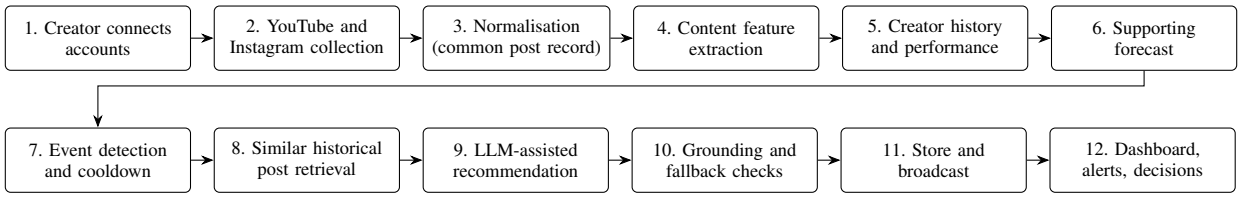


Fig. 1. End-to-end workflow of the system. Steps 1–6 prepare a content-aware, creator-aware post record; steps 7–12 handle events and recommendations. Forecasting (step 6) is one supporting input.

TABLE II
COMMON POST RECORD AND WHAT IT PRESERVES

Field	Purpose and platform detail kept
post_id, platform, platform_post_id	Internal id; origin platform and its own identifier
creator_id	Link to the creator's history
content_type	Reel, Short, video, image or other supported type
caption_text, media_reference, duration	Shared content information, where available
published_at	Publishing time
content_features	Extracted text, visual and speech descriptors
engagement_snapshot	Time-specific observations with original metric names
platform_metadata	Original platform-specific fields, unchanged

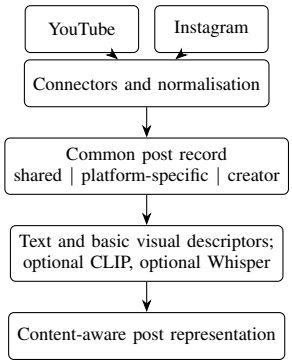


Fig. 2. Cross-platform content representation. Shared content information is standardised; platform-specific fields and creator history stay attached to the record.

F. Creator-aware forecasting (supporting module)

Performance is expressed relative to the creator's own history. With y_i the observed or estimated engagement of post i and b_c the creator's historical baseline,

$$r_i = \frac{y_i - b_c}{b_c}, \tag{1}$$

so $r_i = 0.5$ means 50% above the creator's usual level. How b_c is computed (mean, median, window) is *[CONFIRM: baseline definition in code]*. The forecast gives context to recommendations; it is not claimed to be accurate. Section IV-D explains which implementation produces it.

G. Event detection

Three events are monitored: a new upload, underperformance against an expected or configured trajectory, and

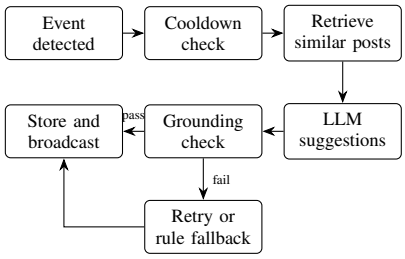


Fig. 3. Event-driven recommendation workflow with a cooldown, a grounding check and a fallback.

a comment surge against configured detection logic. Detection is rule or threshold based and not an anomaly-detection model. The workflow is: receive a platform event or a scheduled synchronisation result; identify the post; fetch its record and snapshots; update the record; check the event conditions; apply a cooldown so the same event does not alert repeatedly; start the recommendation workflow; store and broadcast the result. Because polling is scheduled, the system is near-real-time, not instantaneous. Thresholds and cooldown length are *[CONFIRM: values in code]*.

H. Historical post retrieval

For a post that triggers an event, the system retrieves similar earlier posts of the same creator. If $\mathbf{q}$ and $\mathbf{d}$ are the vector representations of the current and a stored post, a common similarity is

$$s(\mathbf{q}, \mathbf{d}) = \frac{\mathbf{q}^\top \mathbf{d}}{\|\mathbf{q}\| \, \|\mathbf{d}\|}, \tag{2}$$

and the k highest-scoring posts are returned. The measure and k actually used are *[CONFIRM: similarity and k in code]*. Similarity does not show that one post caused another post's result.

I. LLM-assisted recommendation

The language model receives the post information, content descriptors, the forecast and any explanation, the event, the retrieved posts and earlier recommendation context, and writes suggestions such as reviewing the opening, shortening a long caption or checking the call to action (Fig. 3). Output is checked for grounding in the retrieved evidence; on failure the workflow retries or returns rule-based suggestions, and the result is stored. LangGraph runs the workflow when installed, with a manual fallback. Suggestions are advisory, and no gain in views or followers is promised.

J. Dashboard and decision support

The dashboard shows connected accounts, cross-platform posts, content descriptors, engagement observations, forecasts where available, alerts, recommendations, similar earlier posts, a feedback control, and a draft "what-if" page. The what-if page is described by its actual behaviour in Section IV-D and has not been validated as a predictor of future engagement.

TABLE III
IMPLEMENTATION COMPONENTS AND STATUS

Component	Implementation	Status
Back end	Python, FastAPI	Default
Front end	Next.js, React, TypeScript	Default
Database	SQLAlchemy; SQLite or PostgreSQL	Default
Connectors	YouTube, Instagram	Code present
Text features	Descriptors; hash-based embeddings	Default
Visual features	Basic descriptors; CLIP	Default; optional
Speech	Whisper	Optional
Forecast training	LightGBM training and evaluation	Offline
Live forecast	OpenAI-compatible LLM API	Default
Workflow engine	LangGraph, manual fallback	Optional
Event bus	Redis Streams, in-process fallback	Optional
Messaging	REST and WebSockets	Default

IV. System Implementation

Table III lists the components identified in the source code with their status. *Default* means active in the inspected configuration, *optional* means available but not assumed active, and *not integrated* means present but not connected to the live path.

A. Backend, API integration and database

The back end is a FastAPI service in Python. Platform connectors run through scheduled synchronisation and, where configured, push notifications or webhooks. SQLAlchemy maps records to SQLite or PostgreSQL. The stored record types are creators, accounts, posts, engagement snapshots, forecasts, events, recommendations and feedback.

B. Content processing

Basic text and visual descriptors need no deep model and always run. CLIP and Whisper are optional modules; their use must not be read as evidence that all media is processed by deep learning. Missing modalities are handled explicitly.

C. Forecasting component

Two things must be kept apart. First, a LightGBM training and evaluation pipeline exists [14]. Second, the live forecasting service in the inspected code calls an OpenAI-compatible LLM API to produce estimates and explanations. The trained LightGBM model is therefore *not* claimed to power live predictions, and connecting it to the live service is future work. The LLM-based estimates have not been validated against outcomes.

D. Recommendation component

The workflow contains retrieval, grounding checks, retry or fallback handling and result storage (Section III-I). No grounding score is available yet.

TABLE IV
RECORDS IN THE INSPECTED DATABASE SNAPSHOT

Record type	Count
Creators / accounts	1 / 1
Posts	114
Engagement snapshots	3,150
Forecast records	63
Events	112
Recommendations	51
Feedback records	0

TABLE V
EVALUATION PLAN AND MISSING RESULTS

Area	Experiment to run	Result
Content extraction	Count successful and failed cases per module and modality	[ ]
Cross-platform representation	Case study on comparable posts from both platforms; check shared and platform fields	[ ]
Forecasting (LightGBM)	Time-aware split; MAE, RMSE, Spearman versus a simple baseline	[ ]
Recommendations	Manual rubric: linked to post data, relevant history, specific, unsupported statements	[ ]
Events and latency	Events detected, duplicate alerts, event-to-recommendation delay by stage	[ ]

E. Frontend and communication

The Next.js dashboard talks to the back end through REST and receives alerts through WebSockets. Redis Streams carry events when configured, with an in-process fallback otherwise.

V. EXPERIMENTAL SETUP AND RESULTS

A. Dataset and data collection

Table IV gives the record counts of one inspected database snapshot. They show that post, snapshot, forecast, event and recommendation records are stored. They are implementation evidence and not model performance. The account in this snapshot is a YouTube account, so the snapshot says nothing about Instagram or cross-platform behaviour, and zero feedback records means usefulness rated by creators cannot be reported. *[CONFIRM: re-count records before submission; later dashboard views showed different totals, including two Instagram posts]*

B. Evaluation protocol and required experiments

Table V lists the evaluations that are still to be run. Each red cell is a measured value that must come from the experiment named in the second column.

For forecasting, with N test posts, predictions $\hat{y}_i$ and observations y_i,

$$\mathrm{MAE} = \frac{1}{N}\sum_i |\hat{y}_i - y_i|, \quad \mathrm{RMSE} = \sqrt{\frac{1}{N}\sum_i (\hat{y}_i - y_i)^2},$$
$$\rho_S = \mathrm{corr}\big(\mathrm{rank}(\hat{y}), \mathrm{rank}(y)\big). \tag{3}$$

Spearman's ρ_S matters when the aim is to find relatively better posts. Snapshots of one post must not be split across training and test sets, and training data must come earlier in time than test data. Measured values: **[REQUIRED: MAE, RMSE and Spearman after the time-aware evaluation]**.

C. System-level observations

Dashboard views captured on 31 October 2026 showed the following, which describe the prototype's state and are not results. (i) Instagram held two posts and two followers, which is too little for a cross-platform study. (ii) The what-if page returned HTTP 502, reporting that the external model could not produce a valid forecast. (iii) Recent YouTube posts showed "no estimate yet". (iv) Counters displayed that 14% of advice was supported by post data and 75% used backup suggestions; these are display counters, not an evaluation of grounding. (v) A displayed typical delay of 174.8 s after an alert may include polling wait and is not a measured latency of the processing stages. (vi) One engagement rate of 125% appeared for the account with two followers, which shows that follower-based rates are unreliable for tiny accounts.

VI. DISCUSSION AND LIMITATIONS

The prototype shows that the parts of a content-aware, creator-aware workflow can be connected in one system. It does not show that the suggestions are useful, that the forecasts are accurate, or that the representation works across platforms. The main limits are:

- APIs expose different fields and permissions, so metrics are not equal across platforms.
- One creator and mostly one platform limit generalisation.
- Hash-based embeddings and basic descriptors capture limited meaning, and the optional CLIP and Whisper modules may be inactive.
- The live forecast comes from an LLM and is unvalidated, and the LightGBM model is not wired into it.
- LLM suggestions can be incomplete or wrong, retrieved similarity is not causation, and fallback is frequent in the observed build.
- Other factors that affect engagement are not visible to the system, and platform APIs may change.

VII. FUTURE WORK

Priorities are: connect the trained forecaster to the live service and report time-aware results; collect more creators and Instagram data and run the cross-platform case study; improve multimodal representation and missing-modality handling; add creator feedback and a larger recommendation evaluation; add platforms where APIs allow; and measure event latency by stage.

VIII. CONCLUSION

We described a content-aware, cross-platform decision-support prototype for YouTube and Instagram creators: a common post record, content descriptors, creator history, event detection and retrieval-supported LLM suggestions with

grounding checks and a fallback. Forecasting is a supporting input. The available evidence is the stored records of one YouTube-based deployment and observations of the dashboard, so the work reports an implemented design and an evaluation plan, not performance. The planned experiments in Table V are needed before any claim of accuracy, usefulness or cross-platform generalisation.

ACKNOWLEDGMENT

[CONFIRM: guide, laboratory and funding acknowledgments. If an AI writing tool was used for language editing, disclose it here as the target conference requires.]

REFERENCES

[1] D. Li, W. Li, B. Lu, H. Li, S. Ma, G. Krishnan, and J. Wang, "Delving deep into engagement prediction of short videos," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*. Cham, Switzerland: Springer, 2024, pp. 289–306.

[2] Y. Xu, B. Zheng, W. Zhu, H. Pan, Y. Yao, N. Xu, A. Liu, Q. Zhang, and C. Yan, "SMTPD: A new benchmark for temporal prediction of social media popularity," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, 2025, pp. 18 847–18 857.

[3] C. Riis, D. K. Kowalczyk, and L. K. Hansen, "On the limits to multimodal popularity prediction on Instagram – a new robust, efficient and explainable baseline," in *Proc. 13th Int. Conf. Agents Artif. Intell. (ICAART)*, vol. 2. SCITEPRESS, 2021, pp. 1200–1209.

[4] S. D. Roy, T. Mei, W. Zeng, and S. Li, "Towards cross-domain learning for social video popularity prediction," *IEEE Trans. Multimedia*, vol. 15, no. 6, pp. 1255–1267, Oct. 2013.

[5] B. T. Gia, K. Le, T. Do, T.-D. Mai, T. D. Ngo, D.-D. Le, and S. Satoh, "VRAG: Retrieval-augmented video question answering for long-form videos," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2025, pp. 3698–3707.

[6] H. Liu, S. Jiang, F. Duan, Y. Lyu, X. Wang, H. Ge, and C. Liang, "CadenceRAG: Context-aware and dependency-enhanced retrieval augmented generation for holistic video understanding," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, 2025, pp. 3688–3697.

[7] S. Wu, M.-A. Rizoiu, and L. Xie, "Beyond views: Measuring and predicting engagement in online videos," in *Proc. Int. AAAI Conf. Web Social Media (ICWSM)*, vol. 12, no. 1, 2018.

[8] D. Li, S. Ma, H. Hua, W. Li, J. Wang, C. W. Zhou *et al.*, "VQualA 2025 challenge on engagement prediction for short videos: Methods and results," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, Oct. 2025, pp. 3422–3432.

[9] A. Radford, J. W. Kim, C. Hallacy, A. Ramesh, G. Goh, S. Agarwal, G. Sastry, A. Askell, P. Mishkin, J. Clark *et al.*, "Learning transferable visual models from natural language supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 139, 2021, pp. 8748–8763.

[10] A. Radford, J. W. Kim, T. Xu, G. Brockman, C. McLeavey, and I. Sutskever, "Robust speech recognition via large-scale weak supervision," in *Proc. Int. Conf. Mach. Learn. (ICML)*, ser. PMLR, vol. 202, 2023, pp. 28 492–28 518.

[11] R. Bompilwar and S. Koshatwar, "QualiVision: Multi-modal video quality assessment with quality-aware fusion and discriminative learning strategies," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2025, pp. 3469–3478.

[12] F. Guan, Z. Yu, Y. Lu, W. Luo, Y. Gao, X. Li, and Z. Chen, "MMF-QE: Advanced multi-modal fusion for quality assessment and engagement prediction in user-generated short videos," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2025, pp. 3489–3496.

[13] W. Sun, L. Cao, Y. Cao, W. Zhang, W. Wen, K. Zhang, Z. Chen, F. Lu, X. Min, and G. Zhai, "Engagement prediction of short videos with large multimodal models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2025, pp. 3433–3442.

[14] G. Ke, Q. Meng, T. Finley, T. Wang, W. Chen, W. Ma, Q. Ye, and T.-Y. Liu, "LightGBM: A highly efficient gradient boosting decision tree," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 30, 2017.

[15] P. Lewis, E. Perez, A. Piktus, F. Petroni, V. Karpukhin, N. Goyal, H. Küttler, M. Lewis, W. tau Yih, T. Rocktäschel *et al.*, "Retrieval-augmented generation for knowledge-intensive NLP tasks," in *Adv. Neural Inf. Process. Syst. (NeurIPS)*, vol. 33, 2020.

[16] V. Karpukhin, B. Oguz, S. Min, P. Lewis, L. Wu, S. Edunov, D. Chen, and W. tau Yih, "Dense passage retrieval for open-domain question answering," in *Proc. Conf. Empirical Methods Natural Lang. Process. (EMNLP)*, 2020, pp. 6769–6781.

[17] K. Dela Rosa, "Smart routing for multimodal video retrieval: When to search what," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. Workshops (ICCVW)*, 2025, pp. 4177–4185.

[18] V. Pipoli, A. Saporita, F. Bolelli, M. Cornia, L. Baraldi, C. Grana, R. Cucchiara, and E. Ficarra, "MissRAG: Addressing the missing modality challenge in multimodal large language models," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, 2025, pp. 3215–3224.